

Jayanth Reddy

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SUMMARY

Healthcare Data Analyst with 4+ years of experience in healthcare data processing, statistical analysis, and data visualization using SQL, Python, and business intelligence tools. Expert in cleaning, validating, and analyzing complex healthcare datasets including claims data, EMR systems, and clinical records with comprehensive knowledge of healthcare data standards (HL7, ICD-10, CPT codes) and HIPAA compliance protocols. Proven track record of developing interactive dashboards and reports using Power BI, Tableau, and SQL-based analytics for healthcare providers and payers. Strong background in statistical modeling, trend analysis, and pattern recognition using Python libraries including pandas, scikit-learn, and data visualization frameworks. Demonstrated ability to transform complex healthcare data into actionable insights that improve patient outcomes and operational efficiency, with extensive experience in data quality monitoring, validation methodologies, and healthcare analytics infrastructure. Expert in remote collaboration and cross-functional team environments, with specialized knowledge in patient journey modeling, HCP targeting, and clinical data processing for healthcare organizations.

PROJECTS

Healthcare Claims Data Analytics Platform | *Python, SQL, Power BI, Snowflake, HIPAA Compliance*

- Developed comprehensive healthcare claims data processing pipeline using Python pandas and SQL for analyzing multi-million record datasets from healthcare payers and providers. Built automated data validation and quality monitoring systems ensuring HIPAA compliance and data integrity across EMR, claims, and clinical datasets. Implemented statistical analysis workflows using ICD-10 and CPT code standardization, achieving 98% data accuracy and reducing processing time by 65% for healthcare analytics teams.
- Created interactive Power BI dashboards and Tableau visualizations for healthcare stakeholders, enabling real-time monitoring of patient outcomes, provider performance, and operational efficiency metrics. Applied advanced statistical modeling techniques including trend analysis, pattern recognition, and predictive analytics to identify healthcare utilization patterns and cost optimization opportunities. Delivered actionable insights that improved patient care coordination by 40% and reduced administrative costs by 25% for participating healthcare organizations.

Patient Journey Analytics and EMR Data Integration | *Python, R, SQL Server, Tableau, HL7 Standards*

- Architected patient journey modeling system using Python and R for comprehensive analysis of clinical pathways and treatment outcomes across multiple healthcare providers. Built robust EMR data integration framework supporting HL7 FHIR standards and HIPAA-compliant data processing workflows for longitudinal patient analytics. Developed automated anomaly detection algorithms identifying care gaps and improving clinical decision support systems, processing over 500K patient records with 99.5% accuracy.
- Implemented advanced healthcare analytics solutions using SQL Server and cloud data platforms for real-time patient monitoring and population health management. Created comprehensive reporting frameworks supporting healthcare quality measures, readmission analytics, and provider performance assessment using statistical validation methodologies. Generated evidence-based insights that enhanced care quality metrics by 35% and reduced patient readmission rates by 20% through data-driven clinical pathway optimization and predictive risk stratification models.

EXPERIENCE

Walmart

March 2024 – Present

Senior Healthcare Data Analyst - Analytics & Business Intelligence

Bentonville, AR

- Lead development of advanced healthcare analytics solutions using Python data science stack (pandas, numpy, scikit-learn) and SQL for large-scale healthcare data processing and population health analytics across multi-million member datasets. Architect cloud-native healthcare data platforms leveraging AWS and Snowflake for real-time EMR data processing and clinical analytics infrastructure. Design and deploy HIPAA-compliant data visualization solutions using Power BI and Tableau, implementing comprehensive monitoring and automated reporting for healthcare operations teams.
- Build sophisticated healthcare data pipelines for claims processing, patient journey analytics, and clinical outcomes measurement using advanced statistical modeling and machine learning techniques. Develop automated healthcare reporting systems with strong emphasis on data quality validation, HIPAA compliance, and integration with EMR systems supporting HL7 standards. Create comprehensive healthcare analytics frameworks supporting provider performance analysis and patient outcome optimization, achieving 99.9% system reliability and 60% improvement in operational efficiency across healthcare delivery networks.
- Champion implementation of advanced healthcare analytics methodologies and business intelligence solutions, leading cross-functional teams in developing scalable healthcare data platforms from requirements gathering to production deployment. Drive innovation in healthcare data governance through rigorous validation protocols, automated testing, and continuous improvement practices for clinical analytics infrastructure. Collaborate with healthcare professionals and administrators to translate clinical hypotheses into robust, measurable healthcare analytics solutions, directly influencing patient care quality and operational performance across healthcare organizations.

Mass Mutual

Aug. 2022 – Aug. 2023

Healthcare Data Analyst - Statistical Modeling & Compliance

Hyderabad, IND

- Developed and maintained healthcare analytics models using Python and SQL for clinical data analysis and healthcare operations optimization, implementing statistical modeling solutions for patient outcome prediction, provider performance assessment, and healthcare cost analysis. Built comprehensive healthcare data validation frameworks supporting HIPAA compliance requirements and clinical data quality assurance, applying advanced statistical methods including time-series analysis, regression modeling, and predictive analytics for healthcare data processing and insights generation.
- Implemented automated healthcare data analysis systems using statistical modeling and cloud data platforms, creating interactive dashboards for healthcare administrators and clinical decision support. Applied advanced healthcare analytics methodologies and data science techniques including patient cohort analysis and multi-variate clinical modeling to design efficient healthcare data processing workflows. Established data quality monitoring procedures and HIPAA compliance validation methodologies essential for production healthcare analytics systems and real-time clinical data platforms.
- Participated in cross-functional healthcare analytics initiatives and agile development practices, building expertise in clinical data standards and automated healthcare reporting pipeline development. Supported multiple concurrent healthcare analytics projects using version control systems and statistical validation frameworks, developing skills in stakeholder communication and technical solution design essential for senior healthcare analyst roles. Contributed to healthcare data governance and regulatory compliance initiatives, establishing competencies in HIPAA requirements and clinical data quality assurance essential for production healthcare analytics platforms.

Mindtree

Jan. 2021 – June 2022

Healthcare Data Specialist - EMR Analytics & Reporting

Hyderabad, IND

- Built foundational expertise in Python healthcare data processing and statistical analysis for clinical analytics and healthcare reporting applications, contributing to healthcare data quality initiatives and automated clinical reporting platform development for healthcare services clients. Participated in healthcare analytics projects focused on EMR data processing, statistical validation, and automated healthcare reporting systems, developing core competencies in clinical data processing, healthcare database solutions, and medical data engineering methodologies.
- Supported development of automated healthcare analytics systems and clinical measurement frameworks using Python data stack and SQL databases, gaining experience with healthcare data visualization, HIPAA compliance protocols, and real-time clinical data processing. Participated in collaborative healthcare development projects using statistical validation methodologies and healthcare data standards, learning production analytics practices and technical solution design approaches essential for healthcare data analysis and clinical reporting optimization.
- Contributed to healthcare data quality initiatives and clinical analytics validation methodologies, establishing core competencies in EMR data processing, healthcare reporting, and automated clinical analytics platform development. Built essential skills in production Python development for healthcare applications, statistical analysis, and medical data analytics through structured learning and hands-on healthcare project experience. Developed foundation in healthcare data standards, clinical analytics best practices, and automated healthcare reporting systems crucial for progression in healthcare data analysis and clinical analytics roles.

TECHNICAL SKILLS

Programming Languages: Python (Primary), SQL, R, Java, JavaScript, TypeScript, Bash/Shell Scripting

Data Science & ML: scikit-learn, pandas, polars, NumPy, TensorFlow, PyTorch, matplotlib, Statistical Modeling, Time-Series Analysis

Healthcare Analytics: Claims Data Analysis, EMR Data Processing, Patient Journey Modeling, Clinical Outcomes Analysis, Population Health Analytics

Healthcare Data Standards: HL7, FHIR, ICD-10, CPT Codes, SNOMED CT, Healthcare Data Interoperability, HIPAA Compliance

Business Intelligence & Visualization: Power BI, Tableau, Looker, Interactive Dashboards, Automated Reporting, Healthcare KPI Development

Statistical Analysis: Predictive Modeling, Regression Analysis, Trend Analysis, Anomaly Detection, Clinical Statistics, Multivariate Analysis

Data Quality & Validation: Data Cleansing, Statistical Validation, Healthcare Data Integrity, Quality Monitoring, Data Profiling

Production ML & MLOps: Docker, Kubernetes, Model Deployment, CI/CD Pipelines, Version Control (Git), Automated Testing, MLflow, Weights & Biases

Data Engineering & Pipelines: Airflow, Snowflake, Feature Stores, Data Quality Monitoring, ETL/ELT, Healthcare Data Processing

Big Data Frameworks: Databricks, Apache Spark, Dask, Snowpark, Kafka, Distributed Computing

Quantitative Finance: Systematic Trading, Backtesting Frameworks, Portfolio Optimization, Risk Management, Futures & Options Analytics

Cybersecurity & Network Analytics: PCAP Analysis, Network Monitoring, CVE Assessment, Threat Detection, Vulnerability Analysis

Task Scheduling & Web Frameworks: Airflow, Celery, Prefect, Flask, Django, RESTful APIs, React, TypeScript

Trading & Market Data: Event-Driven Simulations, Walk-Forward Validation, Slippage Modeling, Alternative Datasets, Performance Attribution

Healthcare Analytics: Healthcare Data Analysis, Patient Journey Modeling, HCP Targeting, Clinical Data Processing

Programmatic & AdTech: Audience Segmentation, Identity Resolution, Measurement Platforms, Programmatic Analytics
Statistical Methods: Predictive Modeling, Classification, Clustering, Multivariate Analysis, A/B Testing, Trend Analysis
Cloud Data Platform: AWS (Primary), Azure, GCP, AWS Data Pipeline, RDS, Synapse Analytics, S3, EC2
Databases & Tools: Snowflake, SQL Server, PostgreSQL, MongoDB, Elasticsearch, Graph Databases, NoSQL, Excel (VBA, Power Query), Alteryx
Data Governance: Data Quality Monitoring, Statistical Validation, Regulatory Compliance, Security Protocols, HIPAA Compliance

EDUCATION

University Of Central Missouri

Master's in Computer Science

Vardhaman College Of Engineering

Bachelor of Engineering